Marmara University
Electrical and Electronics Engineering Department
Spring 2023
EE 4091 - Photonics
Assignment (2) - Due April 28, 2023

The Theory of Light

Part I: Wave optics

1) Consider a telescope with lenses of 8 cm diameter. Calculate the diffraction limit for this telescope. Using this telescope, would it be possible to resolve two objects 10 cm apart and 500 meters away from the telescope? Explain. Take average visible light wavelength as 500 nm.

2) Two monochromatic blue in-phase spherical light sources ($\lambda = 460\ nm$), 100 μm apart are projected on a screen 5 meters away. Calculate the location of the first three bright spots on the screen.

Part II: EM Wave optics

3) Refer to the air-glass interface case discussed in class. Considering both TE and TM polarizations, produce the reflection curves ($|\Gamma|$ vs. θ_i) for the following cases. Use MATLAB.
   a) Air-seawater interface ($n = 1.4$).
   b) GaN ($n = 3.6$)-air interface.

4) A TE-polarized light wave is incident from air on a thick glass slab of refractive index 1.8, with angle of incidence θ_i. Plot the normalized reflected power from, and transmitted power through, the air–glass interface versus incident angle. Comment on the curves. What should the incidence angle be to transmit 95% of the power through this interface?

5) A light beam, with polarization direction at 30° from vertical and intensity of 800 W/m^2, passes through a sequence of three polarizers. Measured from the vertical, the transmission axes of the polarizers are at angles of 0°, 30°, and 75°, respectively. What is the intensity of the light when it emerges from each polarizer?

6) The visible light complex refractive index curve of GaAs can be found in the following website (https://refractiveindex.info/).
   a) Find the value of the complex refractive index of GaAs at green color energy.
   b) Calculate the speed of green light inside GaAs.
   c) Calculate the absorption coefficient of GaAs for green color light

Part III: Quantum optics

7) Estimate the ratio of the radiation pressure force by sunlight on a satellite of mass m and cross sectional area A, to the gravitational force by Earth's gravity on the same satellite. Take the sun light intensity as 1.35 kW/m^2. Comment on the results.

8) With a certain metal plate, shining a beam of red light on the metal causes electrons to be emitted. (a) If we replace the red light by blue light, do we know that electrons will be emitted? (b) If the two beams have the same intensity and are incident on equal areas of the plate, do we get the same number of electrons emitted per second in the two cases? Explain. (c) The plate material is most probably Cs (W=1.85 eV), Na (W=2.4 eV) or Ca (W=2.9 eV). Choose one and explain.

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified